

Adaptive Staircase Procedure

PSY310: Lab in Psychology

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GitHub Link

<https://github.com/HetanshJalan/Psychology-Experiments>

Introduction

In psychophysics, the adaptive staircase procedure is a technique for determining the threshold (or just-noticeable difference) for a particular stimulus by altering stimulus levels based on the participant's responses. Here the stimulus was a sinusoidal grating which was tilted to the left or right, and I indicated the direction of the tilt. This type of response-dependent procedure focuses trials at that level where discrimination is difficult, not all intensities at equal levels (Levitt 1971).

Just-noticeable difference (JND) is the smallest difference between two stimuli that can be detected. A 3-down/1-up staircase is typically employed to get to a performance of around 80% correct (PsychoPy, 2026a). Responses were recorded and stimulus was controlled using PsychoPy (Peirce et al., 2019).

Aim- My difference threshold in terms of orientation discrimination was estimated using an adaptive staircase protocol with 3 down/1 up steps.

Hypothesis- The more I lower the tilt, the more difficult it should be to discriminate and the more stable the staircase should be on a small range near to my threshold.

Method

Participant

The experiment was conducted by 1 female and myself (both 20 years old) as a student and student researcher. Data were collected with the pilot label.

Materials

I created and ran the experiment in PsychoPy 2026.2.3. The computer was equipped with the processor from the 11th generation of Intel Core i5-1135G7 (2.40 GHz), 16 GB of RAM, NVIDIA GeForce MX350 (2 GB), and Windows 11 Home Single Language version 25H2. The refresh rate of the display was around 59.92 Hz as recorded. Responses were collected using the keyboard.

I employed a Gaussian-masked sinusoidal grating which was 0.3×0.3 PsychoPy height units and had a spatial frequency of 8 cycles per stimulus unit. The grating was presented at both positive and negative values of tilt. The fixation cross was displayed for 500ms, followed by the grating for 300ms. I responded with the left or right arrow key.



Figure 2. Left-oriented tilted grating stimulus.

Figure 1- Example of the tilted sinusoidal grating used in the task.

Staircase parameters

Parameter	Setting
Starting value	10° tilt magnitude
Minimum / maximum	1° / 20°
Step type	Linear
Step sizes	2°, 1°, 1°, 0.5°
N up	1 incorrect response

N down	3 correct responses
Observations	100 trials
Response mapping	Left = -tilt; Right = +tilt

Table 1- Parameters used for the 100-trial staircase.

Procedure

I was requested to make the quickest and most accurate judgement about the direction of the tilt of the grating. Every trial was started with a 500ms fixation cross, followed by a 300ms sinusoidal grating. The direction of the orientation was reversed in different trials: left response to negative tilt, and right response to positive tilt. The staircase was then incremented (or decremented) the next tilt amount according to the 3-down/1-up rule in the programmed sequence after each answer. I completed 100 trials.

Data treatment

I have reviewed the exported csv data for accuracy of responses, magnitude of tilt, intensity of the stairs, and reaction time. I was converting the reaction-times from seconds to milliseconds. Implementing the instructions in the experiment, I found the 5 last 1 to 0 reversals (one correct response after an incorrect response) and calculated the mean staircase intensity for the last 5 reversals to get an approximation of the difference-threshold. The analysis is descriptive only because it was based on one participant.

Results

I made all the 100 trials. I got 94 of them right, and 6 wrong, which was an overall accuracy of 94.0%. Mean reaction time was 589.2 ms (SD = 313.0), with a median of 549.2 ms.

After applying the 1-0 reversal rule, the last five intensities of the reversals were 1.0°, 2.0°, 1.0°, 1.0°, and 1.0°. The difference threshold for this run was 1.20° which is their mean.

Table 2. Final five 1-to-0 reversals

Reversal	Trial	Tilt (°)	Staircase intensity (°)
1	7	-1.0	1.0
2	9	2.0	2.0
3	32	1.0	1.0
4	73	-1.0	1.0
5	92	-1.0	1.0
Mean			1.20

Table 2- The final five 1-to-0 reversals used to calculate the measured difference threshold.

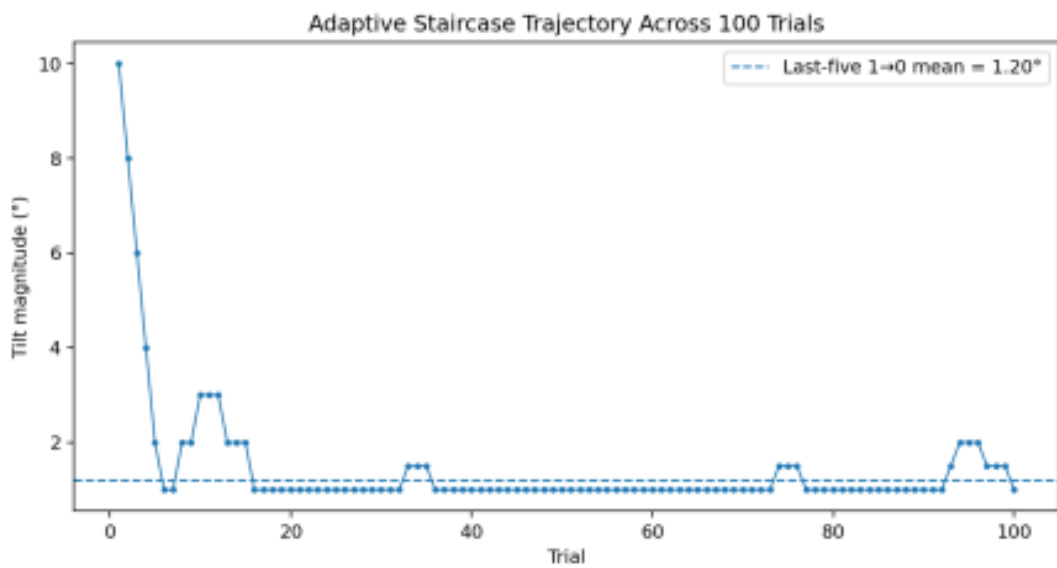


Figure 1 - Tilt magnitude across the 100 staircase trials. The dashed line marks the mean of the final five 1-to-0 reversals.

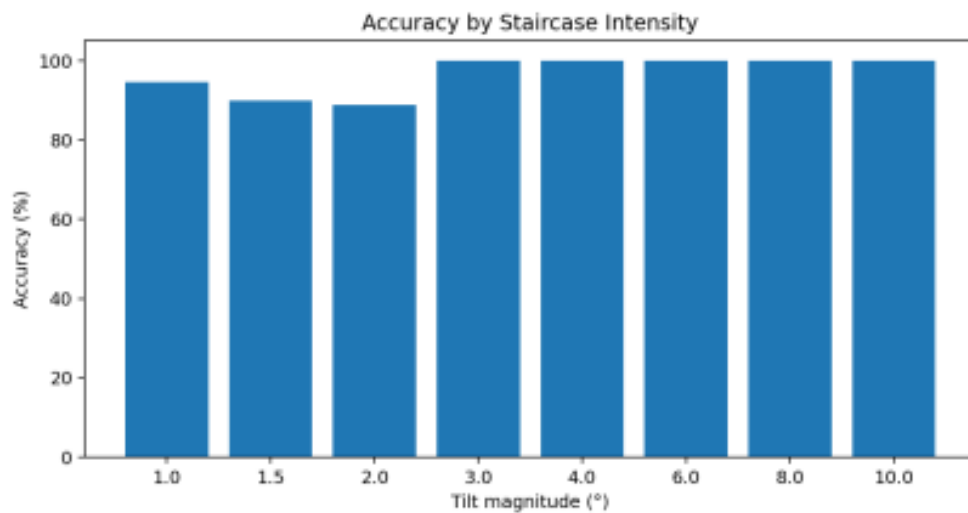


Figure 3 -Accuracy at each tested staircase intensity.

Discussion

The primary outcome was a measured difference threshold of 1.20° for the last 5 reversals 1-0. The staircase ramped up rapidly from the starting angle of 10° towards a smaller angle and remained close to the lower end of the programmed range during most of the remaining trials. This means that the procedure could adjust the stimulus to my responses.

The threshold value is the magnitude of tilt around which the last time the final reversal criterion has been met in this specific simulation. It provides an indication of how sensitive I am to small orientation differences in that stimulus, at that time and under those display conditions. Not to use as a population threshold as a single participant was used for the task.

Overall accuracy with 94% for 100 trials. The high accuracy is to be interpreted in conjunction with the adaptive design: the staircase did not sample all possible tilt values uniformly and the majority of trials were at 1° or 1.5° . The reaction time was secondary to the threshold calculation and was only used as a descriptive measure. The 100 trial run is a more reliable basis for the calculation of the last 5 values of the reversal than the previous 50 trial run, but more trials and participants are required to assess reliability and generalizability.

Conclusion

The 100-trial adaptive staircase resulted in an estimate of the difference-threshold of 1.20° . The outcome is the orientation difference that the smallest staircase minimizes towards this participant in the conditions of the experiment.

References

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